Seeing is Deceiving: An ethical framework for Navigating the AI generated content

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1. ***Abstract***—

The accelerated evolution of generative AI technologies has led to an exponential increase in synthetic digital content, amplifying concerns related to misinformation, manipulation, and content authenticity. In response to these emerging challenges, this study introduces a theoretically grounded and multi-layered framework designed to ensure provenance and trust in AI-generated imagery. Our approach leverages a fusion of cryptographic hashing, blockchain technology, steganography, and metadata embedding to enable reliable origin tracing of synthetic media. Inspired by the traceability mechanisms employed in blockchain-backed assets such as NFTs, each AI-generated image is assigned a unique, tamper-evident hash derived from its inherent visual (or audio) features in combination with user-specific metadata. This hash is securely stored on a distributed blockchain ledger, while the corresponding block address is both encrypted and embedded directly into the content using metadata tags and steganographic techniques.

This dual-layered embedding ensures resilience: even in cases where metadata is stripped—commonly observed during social media redistribution—the hidden steganographic signature remains retrievable for validation. The inherent one-way characteristic of cryptographic hash functions guarantees that even minor alterations to the original content result in substantial hash deviations, thus reinforcing the integrity of the verification process. The proposed framework further envisions a government-authorized verification portal, which would serve as a centralized interface for content authentication. Through this portal, users can verify block-level provenance, authenticate content origins, and trace the content creator's identity using linked IP addresses and session metadata. While this system remains conceptual, it offers a forward-looking model that emphasizes transparency, accountability, and digital trust in increasingly AI-driven ecosystems. It represents a foundational step toward establishing a secure infrastructure for the authentication and traceability of synthetic media in the public domain.

2. Introduction

Advancements in artificial intelligence have radically transformed content creation—particularly in generating photorealistic images—making them nearly indistinguishable from those produced by human hands. While this capability drives innovation, it simultaneously introduces serious challenges in distinguishing authentic media from synthetic, thereby fueling misinformation, identity fraud, and unethical content manipulation. Traditional methods of verifying digital content, such as C2PA metadata frameworks[1] or manual inspection, are rapidly becoming ineffective. Metadata can be stripped or altered, and AI-generated images can bypass even expert visual scrutiny. In response, we propose a multi-layered verification framework that assigns each digital image a robust and unique fingerprint at the point of creation.

This fingerprint is derived from the image's core properties, user-specific data (like IP address and session ID), and processed through a cryptographic hash function. The output is stored on a blockchain to ensure immutability and decentralization. The resulting block address is then encrypted and embedded in two redundant forms: as metadata and through steganographic encoding within the image pixels. Even if one layer is removed (e.g., by social media compression or metadata wiping), the other layer provides recoverability. Our framework categorizes content as either:

- Verified (Traceable): Content with intact or recoverable embedded fingerprints, allowing full traceability to its generator.

- Unverified (Traceability not confirmed): Content lacking valid fingerprints, flagged as potentially manipulated or unauthenticated.

To facilitate public verification, we propose a government-regulated portal that allows users to upload any image and instantly verify its authenticity by comparing extracted data against the blockchain record. If metadata or steganographic values are found, the system

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retrieves the original block address and reconstructs the hash for verification. If discrepancies are found, each layer is individually checked against the database, ensuring no legitimate content is wrongly discarded.

While this framework is currently theoretical, it draws inspiration from real-world implementations such as NFT-based ownership verification, proving the practicality of embedding content identifiers on decentralized ledgers. It also leverages the one-way, collision-resistant nature of hash functions, ensuring that even minor changes to the image result in drastically different fingerprints—thereby making forgery detectable. Furthermore, it acknowledges the fragility of steganography to transformations like cropping or filtering and highlights metadata's vulnerability to deletion by platforms for storage optimization.

This paper outlines the architecture, methodology, potential real-world applications, and ethical implications of implementing such a system. By encouraging widespread adoption, we aim to foster a digital ecosystem rooted in trust, accountability, and content transparency.

3. Literature Review

With the rapid advancement of generative artificial intelligence, digital content creation has entered a new phase where synthetic images and media often appear indistinguishable from real ones. This shift has raised serious concerns about the spread of false information, identity theft, and manipulation of digital content. To counter these challenges, researchers and developers have proposed various solutions, many of which revolve around blockchain technology, cryptographic verification, and embedded metadata.

One of the most recognized efforts in this area is the C2PA standard introduced by Rosenthal[1]. It focuses on attaching detailed metadata to digital content to help trace its origin. However, the approach depends heavily on platforms preserving this metadata, which is often removed during compression or social media sharing. In a similar direction, Han and his team[2] suggested a trust-based authorization model for verifying content on social platforms. While useful, such models still struggle with maintaining integrity once the data moves outside controlled environments.

Blockchain has been explored as a tool to bring more transparency and reliability to online information. Research by Seneviratne[3] and by Shae and Tsai[4] supports the idea that the permanent and secure nature of blockchain records can be used to verify the authenticity of online content. These studies present blockchain as a trustworthy method for keeping a history of content ownership and changes, which could help fight against manipulated or fake media. In a more practical context, researchers like Jain[5] and Gaikwad[6] have demonstrated how blockchain can secure academic records and transcripts. Their work shows that blockchain can help verify sensitive data and ensure it remains unchanged. Although these use cases focus on documents, the concept of traceability and verification is applicable to digital media as well.

To address the limitations of relying solely on metadata, researchers have turned to steganography. Anderson and Petitcolas[7] examined its strengths and weaknesses, noting that while it can be effective for hiding data inside images, it is not always reliable after the image is edited or compressed. Morkel and his colleagues explored techniques like hiding information in pixel values or using mathematical transformations within images.[8] These methods offer a creative way to embed verification information, though they also face challenges when media is altered. Despite these advancements, many existing solutions[9] are limited by their dependence on centralized systems, vulnerability to editing, or lack of flexibility across different media types. The framework proposed in this study builds upon these earlier efforts by combining multiple strategies into one cohesive system. It integrates secure hashing, embedded identifiers, and blockchain storage to create a multi-layered structure that can verify content even after it has been shared or modified.

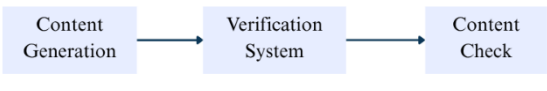
The paper by Elkhatat, and others[10] depict how detection by AI can result in flagging the real too. We believe that watermarking approaches like WEvade[11] and SoK[12] prove to be more robust, even against many planned attacks.

What makes this approach unique is its focus on redundancy and resilience. By embedding encrypted content information both visibly through metadata and invisibly through steganography, it ensures that at least one form of verification remains intact. The system also includes a public verification portal that allows users to check whether content is original or has been tampered with. This combination of technologies creates a more practical and trustworthy model for confirming the origin of digital media.

Overall, this framework moves beyond traditional verification techniques by creating a secure and adaptable solution that addresses the current gaps in content authentication. It supports a future where digital trust, content integrity, and ethical use of artificial intelligence are not just ideals but achievable standards.

4. Proposed Methodology

This research focuses on developing a secure and robust



framework for tracking AI-generated content using cryptographic hashing, steganography, and blockchain technology. The methodology consists of several phases, including data collection, hash generation, embedding mechanisms, verification mechanisms, and adaptation for different content types.

4.1 Data Collection and Feature Extraction

To establish a secure and verifiable digital signature for AI-generated content, a unique identifier (denoted as **V**) must be generated based on a robust set of extracted features. These features vary across content types—image, video, audio, and text—and are organized into two primary categories: user-specific metadata and content-specific descriptors.

4.1.1 User Identification

To ensure traceability and accountability, the following user and platform-level metadata are collected:

- **Network and Device Metadata**: Includes the user's IP address, device identifier (Device ID), session ID, and timestamp of content generation.

- **Platform-Specific Metadata**: Captures the source organization or platform (e.g., website, mobile application), along with its unique digital signature, application header, and cryptographic key. These attributes collectively verify the authenticity of the content's origin.

4.1.2 Content-Specific Feature Extraction

The extraction process for unique content features varies by media type:

A. Images

- The image is processed at the pixel level.
- For each pixel, the program extracts its RGB values:
  - Each channel (Red, Green, Blue) is represented by 1 byte (8 bits).
  - Thus, each pixel occupies 3 bytes (24 bits) in total.
- A notable challenge arises when the image is uniformly colored (e.g., all white), potentially leading to duplicate or non-unique hash values.
- To address this, collision resolution techniques are incorporated, such as embedding device-specific entropy or minor pixel perturbation during hashing.

B. Videos

- Videos are parsed frame-by-frame, with each frame treated as a standalone image.
- Each frame is analyzed at the pixel level to extract RGB values, similar to the image processing pipeline.
  - Each pixel again contributes 3 bytes (24 bits).
- Video size scales with resolution and frame rate.
- For instance:
  - A video at 1920x1080 resolution and 30 frames per second (fps) produces approximately:
  - $1920 \times 1080 \times 3 \times 30 = 186{,}624{,}000$ bytes/sec $\approx$ 178 MB/sec (uncompressed).
- Redundancy becomes a concern when multiple frames are identical or near-identical (e.g., static scenes), potentially generating repeat hashes.
- This is mitigated by applying frame sampling techniques and collision-resistant hashing algorithms to ensure variability and uniqueness.

C. Audio

- Audio content is analyzed using key acoustic descriptors derived from the waveform and its frequency-domain representation (spectrogram).

- The following features are extracted:
  - Spectral Centroid: Indicates the "center of mass" of the spectrum.
  - Spectral Bandwidth: Measures the range of frequencies present.
  - Spectral Flux: Quantifies changes in spectral content over time.
  - Zero-Crossing Rate: Counts signal sign changes over a time window.
  - RMS Energy: Reflects signal amplitude strength.
  - Duration: Total length of the audio in seconds.

- These features are organized into a sequential array and serve as the input (V) for the hashing function, creating a unique fingerprint for each audio clip.

4.2 Hash Generation an Blockchain Storage

To ensure the authenticity and traceability of AI-generated content, a cryptographic hash function is applied to the Input Content (V). This process generates a unique identifier HV, which serves as a digital fingerprint of the content.

Hashing Algorithm Selection:

The hashing function must be collision-resistant, secure, and efficient to prevent duplication or manipulation of content. Suitable cryptographic hash functions include:

- **SHA-256** is widely used in blockchain systems and digital certificates, offering 256-bit hash outputs and proven resistance against preimage and collision attacks.

- **BLAKE2** – Designed as a faster alternative to SHA-3, BLAKE2 delivers high performance without compromising cryptographic strength. It is optimized for modern processors and supports personalization and keying, making it well-suited for scenarios requiring speed and security.

- **Argon2** – Although primarily a password-hashing algorithm, Argon2 is highly secure due to its resistance to GPU and side-channel attacks, and its configurability in terms of memory usage, execution

time, and parallelism. Its robustness makes it particularly valuable for securing sensitive content where resistance to brute-force attempts is critical.

The hash value is computed as:
HV=Hash(V)
Where:
V = Input Content

Blockchain Integration:

- The hash value (HV), combined with metadata such as IP address, timestamp, and platform header (W), is stored securely in a new block on a private blockchain.
- This process guarantees immutability, verifiability, and distributed traceability.
- The block address (BA) is then generated as a unique pointer to this transaction.

4.3 Encryption and Embedding

4.3.1 Encryption of Block Address

- The block address (BA) is encrypted using a symmetric or asymmetric encryption algorithm (e.g., AES-256 for symmetric or RSA for asymmetric) to prevent reverse tracing or tampering.

- This encrypted value (EncBA) acts as the secure content identifier embedded within the media.

4.3.2 Embedding into Content

To make the content verifiable and tamper-resistant, the EncBA is embedded in two layers:

a) Metadata Embedding

- Images & Videos: Stored within EXIF/IPTC metadata tags

- Audio: Stored in ID3 metadata tags

- Text: Embedded as zero-width characters or watermarks

b) Steganographic Embedding

If metadata is stripped or modified, EncBA can still be retrieved via steganography:

- Images: Least Significant Bit (LSB) Encoding or DCT Coefficient Modification.

- Videos: Frequency domain embedding in prime-numbered frames

- Audio: Frequency or phase-based encoding, Stored in ID3 metadata tags

- Text: Font variation or invisible characters

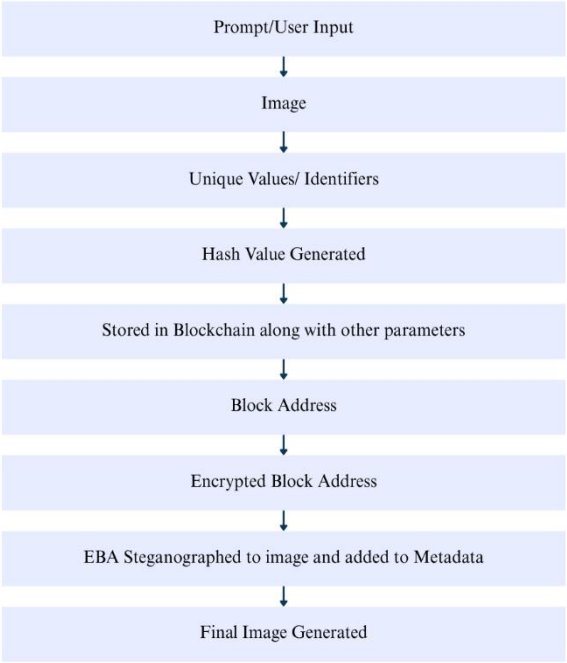
This dual-layer embedding ensures that even if one layer is compromised, the other can be used for verification. By integrating steganographic embedding alongside metadata storage, the system ensures tamper-resistant tracking of AI-generated content across various platforms.

4.4 Output Generation

At this stage:

- The AI-generated image is now embedded with a secure encrypted reference to its blockchain identity
- It contains EncBA in both metadata and steganographic layers
- The content is ready for distribution, use, or sharing with embedded traceability intact.

This structured approach provides a secure, verifiable, and tamper-proof mechanism for tracking AI-generated content across platforms.



4.5 Methodology (Part II): Verification Workflow from Image to Identity

To ensure the authenticity, origin, and integrity of AI-generated images, the proposed framework incorporates a robust verification mechanism that leverages steganographic embedding, metadata analysis, decryption, blockchain querying, and cryptographic hash comparison.

4.5.1 Image Ingestion for Verification

The verification process begins when a user:

- Uploads a suspected/generated image to the verification portal (e.g., via web interface or mobile app)
- Shares the content across public platforms (e.g., social media), where it is picked up by the framework

Each uploaded image is expected to contain a dual-layer embedded value (EncBA):

- Encrypted Block Address in metadata
- Same value embedded via steganography

4.5.2 Dual-Layer Extraction and Consistency Check

Step 1: Extract Embedded Identifiers

- From Metadata (EXIF/IPTC, ID3, etc.)
- From Steganographic Layer (via LSB decoding, DCT coefficient extraction, etc.)

Step 2: Integrity Verification

- Compare both values of EncBA
- If they match, mark them as verified and proceed to decryption
- If they don't match, assume potential tampering and mark it as unverified, then:
    - Decrypt both values separately
    - Attempt verification with both block addresses independently in the next step

Step 3: Decrypt EncBA

- Use the corresponding decryption method (e.g., AES-256 or RSA)
- Retrieve the original Block Address (BA) from EncBA

Step 4: Query the Blockchain

- Use BA to access the stored hash block
- This block contains:
    - Original hashed value (HV)
    - Associated user identifiers (IP, Session ID, Timestamp)
    - Platform/organization tag (W)

Step 5: Recompute Hashed Value

- Apply the same hash function to:
    - Extracted visual features (V) from image (using prime-pixel method).
    - Pass them to hash function and recompute hash. value.

Step 6: Compare Hashes

- Compare HV_recomputed with the original HV retrieved from the blockchain.

- If they match:
    - Image is authentic and untampered
    - Content integrity confirmed
- If they do not match:
    - Either content was modified or not generated via the registered framework.

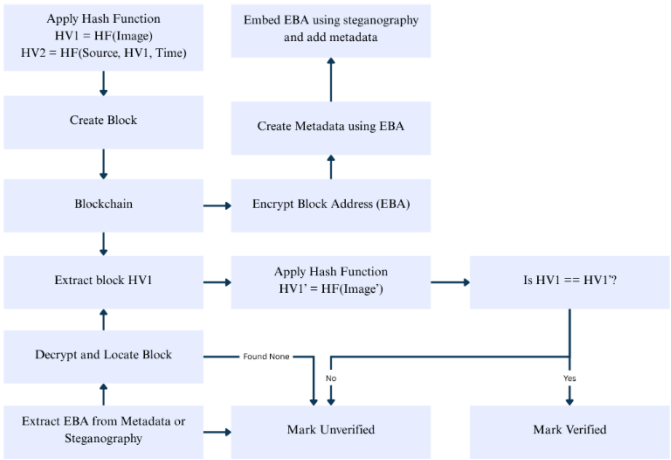
4.6 Final Mapping and Attribution

If the recomputed hash matches the stored one:

- The system confirms the image's authenticity

- Displays or logs the original content generator's details:
    - IP Address
    - Platform Name
    - Session ID
    - Timestamp
    - Any other registered metadata

This verification can be public-facing (for user awareness) or private (for legal/governmental access) based on access privileges.



5. Conclusion

The rapid evolution of artificial intelligence in digital content generation has introduced new challenges in verifying the authenticity and provenance of media. This study presents a multi-layered verification framework that integrates cryptographic hashing, blockchain storage, metadata tagging, and steganographic embedding to ensure content traceability and integrity across platforms.

The proposed approach emphasizes both technical robustness and operational resilience. Cryptographic hashing provides a secure, one-way mechanism to generate unique digital fingerprints, ensuring that any alteration in the content results in a distinctly different output. Metadata embedding allows for straightforward identification of origin, while steganographic techniques serve as a covert and persistent layer of verification. This dual-embedding strategy enhances the framework's fault tolerance, especially in scenarios involving metadata stripping or image transformation during social media redistribution.

Although the system remains conceptual, its foundations are grounded in technologies with demonstrated real-world

viability, including blockchain-backed verification models used in digital ownership and academic credentialing. The inclusion of a publicly accessible verification portal further strengthens its utility by enabling users to independently assess content authenticity through cross-referencing with blockchain records. This allows us to urge the public, in order to trust only verified content, and any mal-practice in a verified content can be traced back to someone being held accountable.

On the other hand, similar to a recent work by Gao and his colleagues in their paper[13], we can further experiment by using Generative AI technologies to increase adoption of our technology.

In summary, the framework proposed in this research offers a scalable and secure solution to the growing problem of synthetic media verification. It provides a forward-thinking model that promotes ethical accountability, content transparency, and digital trust in artificial intelligence-driven ecosystems. By bridging current gaps in media provenance, it contributes meaningfully toward building a resilient and verifiable digital infrastructure.

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